RAJALAKSHMI ENGINEERING COLLEGE

Akash M 250701034-2025-CSE

A2

Started on	Wednesday, 7 October 2026, 10:18 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:20 PM
Time taken	2 mins 22 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main(void) {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int *a = malloc((size_t)n1 * sizeof(int));
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);
```

```
        scanf("%d", &n2);

        int *b = malloc((size_t)n2 * sizeof(int));
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;
        int first = 1;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (!first) printf(" ");
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            } else if (a[i] < b[j]) {
                i++;
            } else {
                j++;
            }
        }

        printf("\n");
        free(a);
        free(b);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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